Suvanth Erranki

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Education

Carnegie Mellon University, School of Computer Science – GPA: 4.0/4.0 Expected May 2029
Bachelor of Science, Computer Science Pittsburgh, PA

Relevant Coursework: Linear Algebra, Imperative Programming, Math Concepts for Computer Science, Real Analysis (at Princeton), Number Theory (at Princeton)

Activities & Experience

USA Computing Olympiad Staff Sep. 2026 – Present

- Assisting with problem-writing, testcase generation and validation, coding solutions, writing editorials, etc. for the upcoming 2026-2027 season of the USA Computing Olympiad.

Software Engineer Intern Jun. 2026 – Aug. 2026
Lockheed Martin *Moorestown, NJ*

- Built a natural-language cost-analysis tool over a 220K+ row aerospace parts dataset, using an LLM to translate user questions into DuckDB SQL and pandas operations to filter and aggregate the data.
- Modeled query results as a DAG of part dependencies and used topological sort to propagate cumulative cost and labor deltas.
- Built an interactive Plotly/Dash dashboard of cost-flow graphs with per-category breakdowns, automatically flagging nodes with >5% weekly Estimate at Complete (EAC) growth for review.

Lead Organizer, West Windsor-Plainsboro Programming Competitions Feb. 2024 – Jun. 2026

- Co-founded a 501(c)(3) hosting programming contests for students, sponsored by Jane Street and HRT; directed 4 contests for 500+ students and distributed $10K+ in prizes and merchandise.
- Wrote contest problems, solutions, and editorials; built the grading infrastructure and judgehosts, and coordinated testing.
- Co-led the first in-person edition at The College of New Jersey. Directed an 8-person team with 2 co-leads, overseeing sponsorships, event logistics, advertising, and school outreach. https://wwppc.org

Awards

USA Computing Olympiad Platinum Division & 2026 Proctored Contest Invitee (one of 75)	Mar. 2026
MIT Battlecode 2025 Finalist – 1st/113 HS Teams, 7th/504 Overall (Livestream | Postmortem)	Jan. 2025
1st Place — Cornell HS Programming Contest (75 teams)	Mar. 2025
1st Place — UPenn PClassic Advanced (50 teams)	Apr. 2025
1st Place — Lockheed Martin Code Quest Advanced (30 teams)	Feb. 2026

Projects

WWP Programming Competitions Judgehost | *TypeScript, Node.js, Linux*

- Sandboxed submissions with the isolate library, enforcing per-language time and memory limits; added custom testcase generators, custom checkers, interactive problems, and Polygon-format problem import.
- Maintained the code for submission queueing across eight grading servers (after our lead developer graduated). Improved error handling and fixed requeue and timeout bugs.
- Returned verdicts for 5K+ contest submissions in C/C++, Java, and Python/PyPy.

Chess Engine "Kinetic Hero" | *C++23* | Source Code

- Generated moves with magic bitboards and precomputed attack tables, reaching 60M nodes/sec on standard perft suites (M2 Pro, single core).
- Implemented Monte Carlo Tree Search with PUCT selection, storing the tree in a preallocated array to avoid allocation at runtime.

Skills

Languages: C++, Python, C, Java, JavaScript, Node.js, TypeScript, Vue, HTML/CSS
Systems & Tools: Linux, Docker, Git, PostgreSQL, MySQL, Bash, Vim
ML/Data: PyTorch, NumPy, pandas, Matplotlib